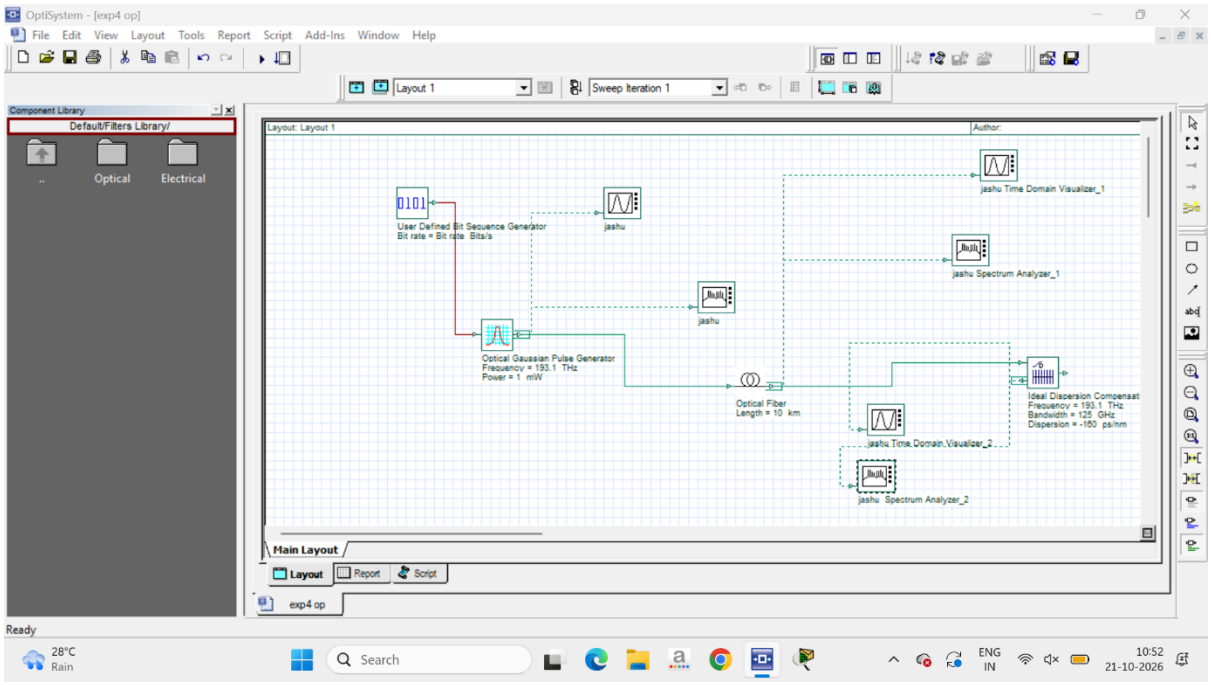
Exp. No: 4

Measurement and Analysis of Dispersion Compensation in Optical Fiber using OptiSystem

OBJECTIVES

1. To demonstrate dispersion compensation using the Ideal Dispersion Compensation component in OptiSystem.
2. To observe the broadening of a 12.5 ps optical Gaussian pulse after propagation through 10 km of single-mode fiber (SMF).
3. To compensate the accumulated dispersion of 160 ps/nm and verify recovery of the original pulse shape.

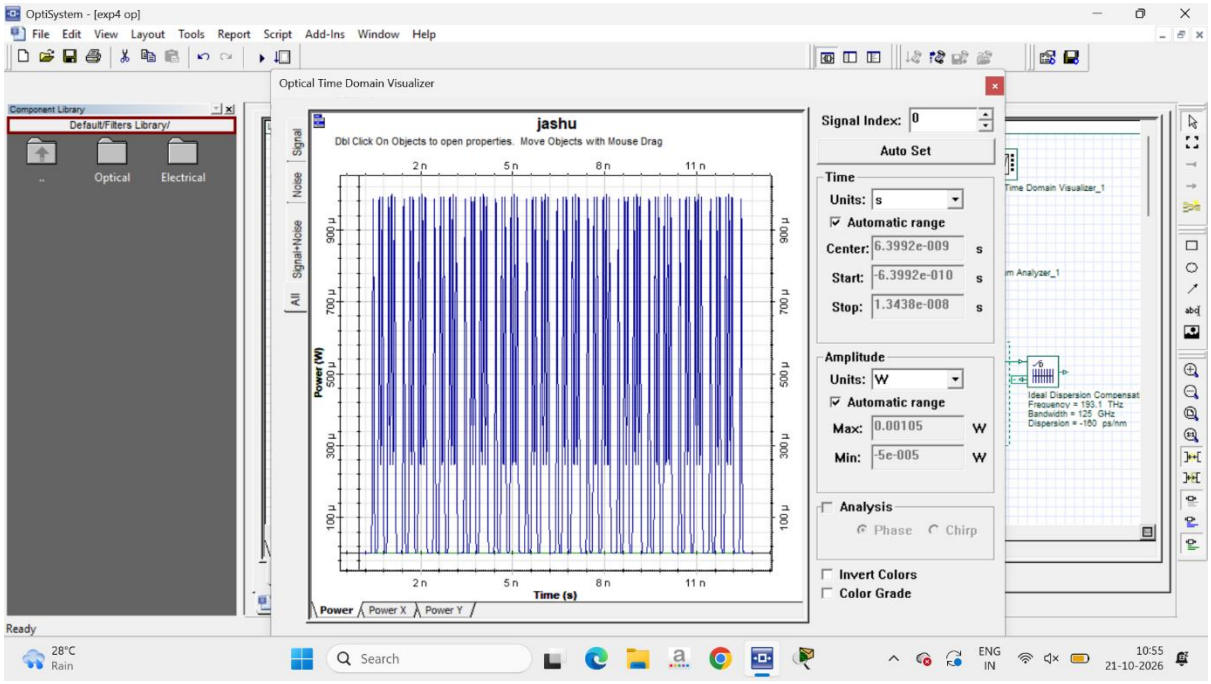
BLOCK DIAGRAM:



OUTPUT GRAPH:

Figure 1: Initial Gaussian pulse

A.


OptiSystem - [exp4 op]
File Edit View Layout Tools Report Script Add-Ins Window Help
Optical Time Domain Visualizer
jashu
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Power (W)
Time (s)
Power
Power X
Power Y
Signal Index: 0
Auto Set
Time
Units: s
Automatic range
Center: 6.3992e-009 s
Start: -6.3992e-010 s
Stop: 1.3438e-008 s
Amplitude
Units: W
Automatic range
Max: 0.00105 W
Min: -5e-005 W
Analysis
Phase
Chirp
Invert Colors
Color Grade
Component Library
Default/Filters Library/
Optical
Electrical
Ready

B.

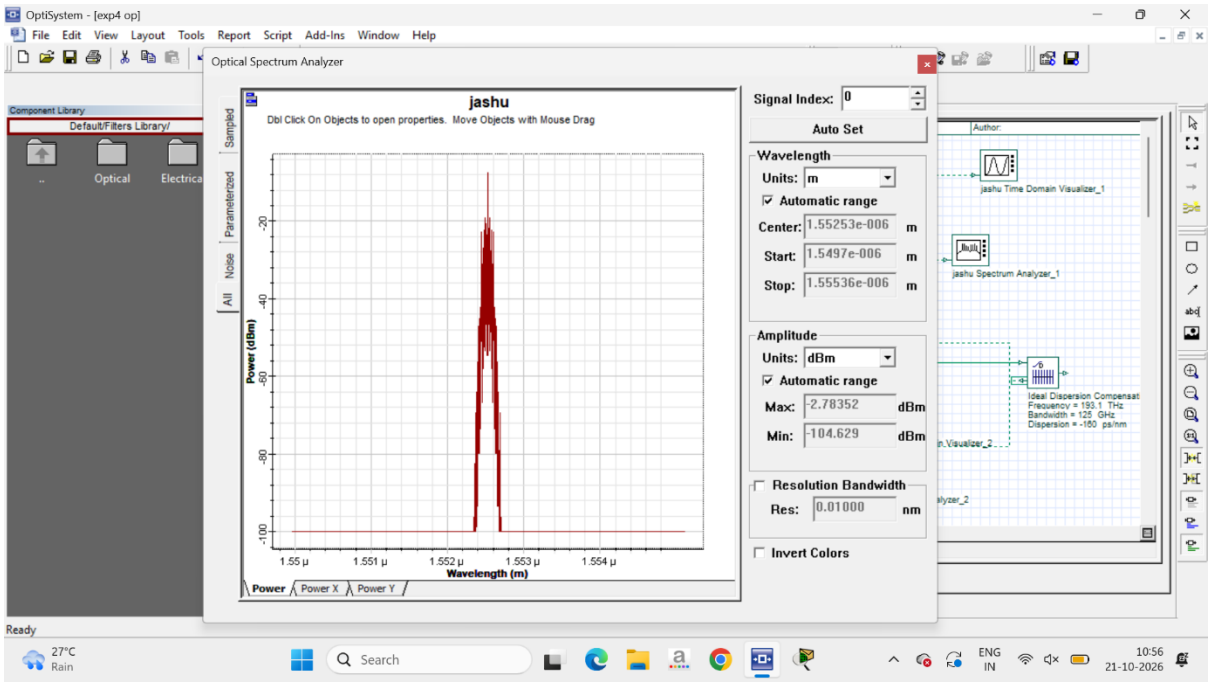
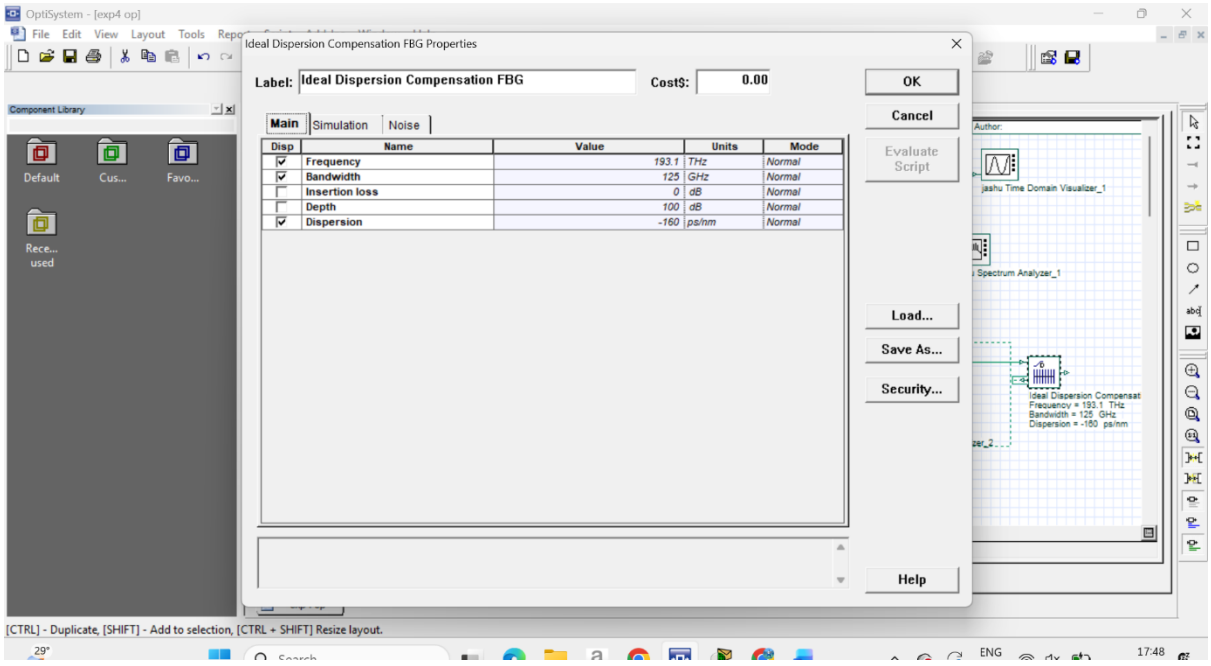
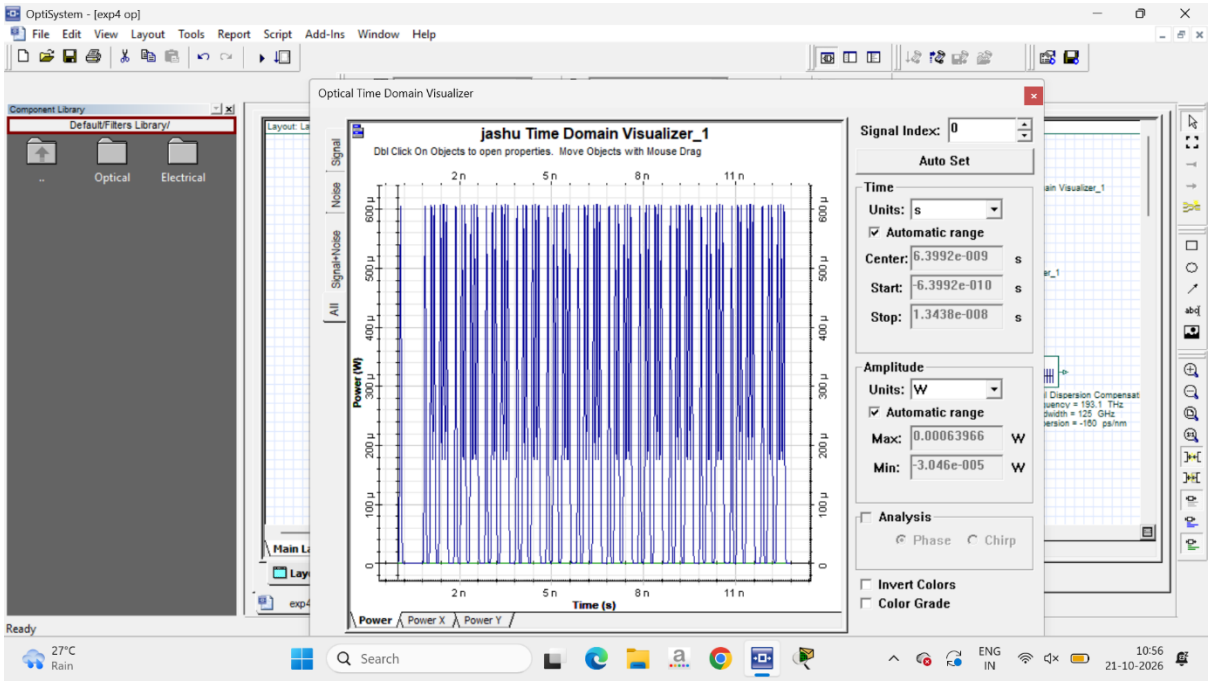

OptiSystem - [exp4 op]
File Edit View Layout Tools Report Script Add-Ins Window Help
Optical Spectrum Analyzer
jashu
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Power (dBm)
Wavelength (m)
Power
Power X
Power Y
Signal Index: 0
Auto Set
Wavelength
Units: m
Automatic range
Center: 1.55253e-006 m
Start: 1.5497e-006 m
Stop: 1.55536e-006 m
Amplitude
Units: dBm
Automatic range
Max: -2.78352 dBm
Min: -104.629 dBm
Resolution Bandwidth
Res: 0.01000 nm
Invert Colors
Component Library
Default/Filters Library/
Optical
Ready

Figure 2: Gaussian pulse after 10 km propagation in SMF

A.



B.



C.

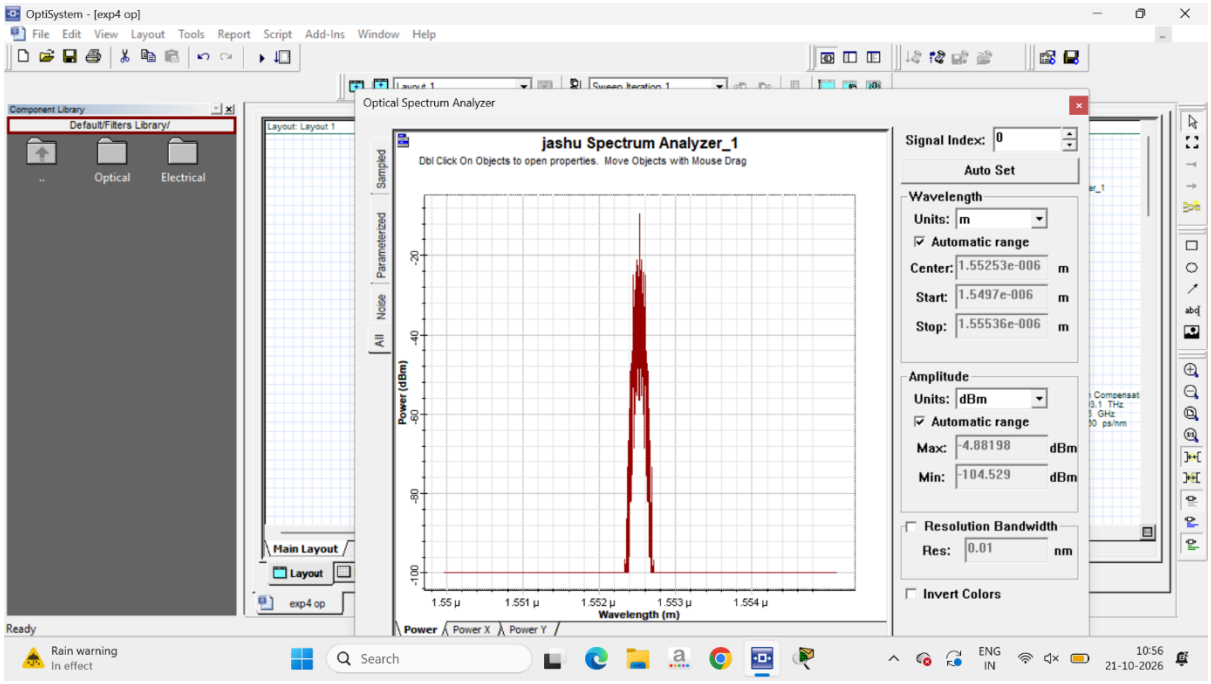


Figure 3: Parameters of Ideal Dispersion Compensation component

A.

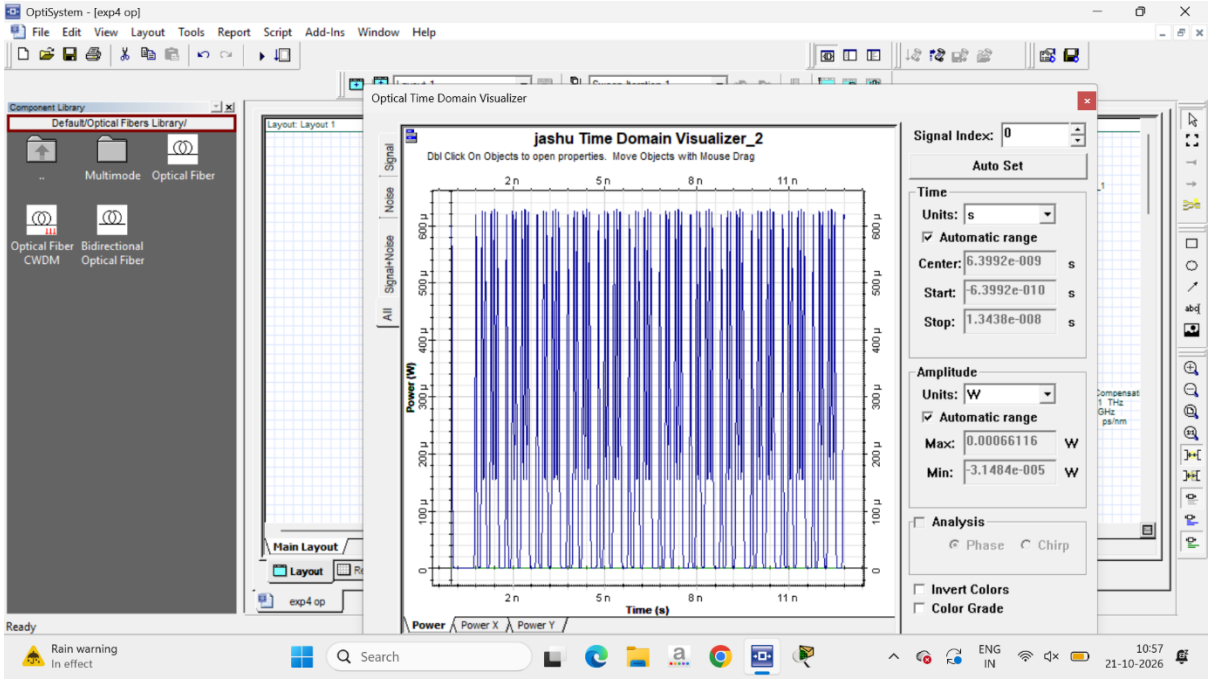
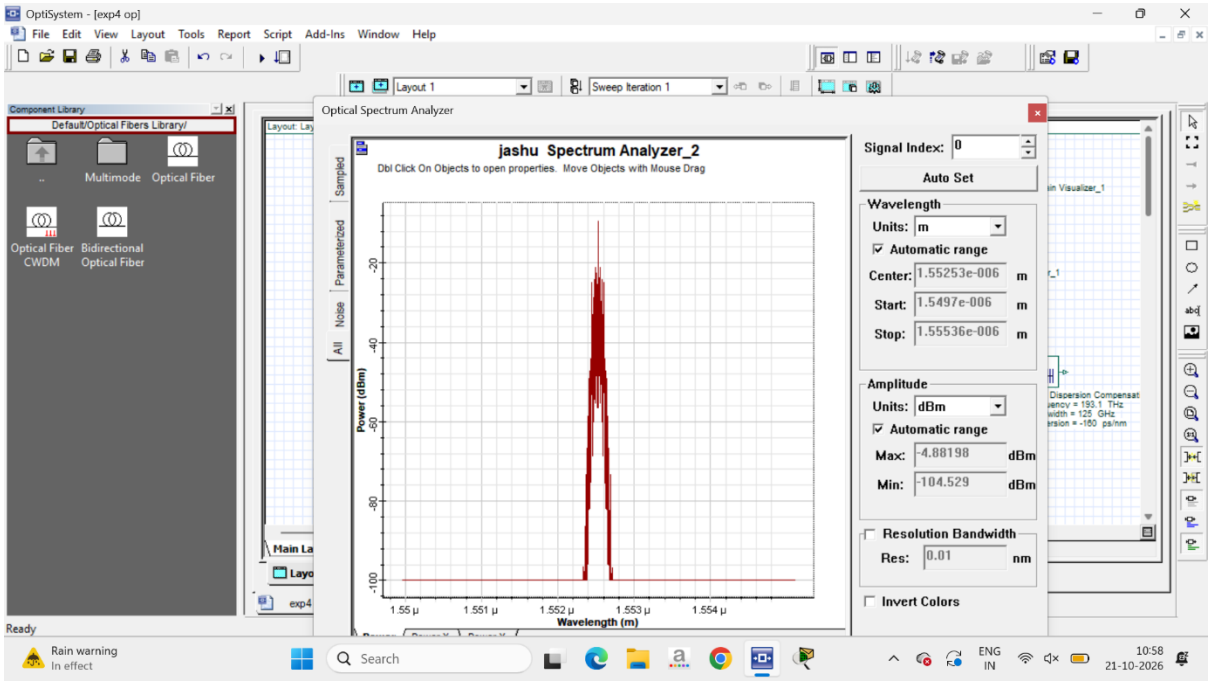


Figure 4: Pulse after dispersion compensation with Ideal Dispersion Compensation component

A.



8. OUTPUT PARAMETERS

Stage	Observation Point	Result	Figure
Initial pulse	Gaussian pulse generator output	Gaussian pulse, 12.5 ps	Figure 1 (A),(B)
After propagation	Output of 10 km SMF	Pulse width increased ≈ 4 times	Figure 2 (A),(B),(C)
Compensator setting	Ideal Dispersion Compensation FBG	Dispersion = –160 ps/nm	Figure 3 (A)
After compensation	Compensator output	Original pulse recovered (exact compensation)	Figure 4 (A)